

Ryan Cho

Senior Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Senior full stack engineer with 8+ years building Java REST APIs, React and TypeScript web clients, and SQL-backed platforms for finance and fintech. Designs schema migrations, query optimizations, and transaction evaluation workflows serving high-volume user flows. Delivers end-to-end features—from interaction design through A/B-tested releases—while mentoring engineers and integrating third-party services at scale.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Led evolution of **Java REST services** powering user onboarding and transaction flows, cutting average API latency by 35% through query tuning and schema indexing across a **PostgreSQL** estate serving 2M+ monthly requests.
- Built **React and TypeScript** web clients end-to-end for income verification and risk dashboards, partnering with design on **interaction patterns** that reduced support tickets by 22% year over year.
- Implemented **monetary transaction evaluation** pipelines integrating third-party credit and identity APIs, hardening validation rules that improved approval accuracy while maintaining sub-200ms p95 response times.
- Optimized **relational database schemas** and migration strategies for portfolio analytics tables, eliminating N+1 query paths and reclaiming 40% faster report generation for internal stakeholders.
- Introduced **A/B testing** infrastructure on feature flags to optimize conversion funnels, informing product decisions that lifted funded-account completions by 12% in controlled rollouts.
- Automated **visual regression** suites in CI for React components, catching UI drift before release and reducing manual QA cycles by 30% across six product squads.
- Containerized services with **Docker and Kubernetes**, standardizing deployment pipelines and enabling zero-downtime releases for payment-adjacent microservices.
- Mentored **three junior engineers** on API design, TypeScript patterns, and SQL performance reviews while driving cross-team architecture discussions with product and operations leaders.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Java REST APIs** for trade settlement and risk reporting modules used by global desks, exposing stable contracts consumed by **React** operations portals across three business units.
- Designed **SQL schemas** and analytical queries for regulatory capital datasets, performing migrations on Oracle and PostgreSQL stores without downtime during peak close periods.
- Integrated **third-party market data APIs** into pricing engines, adding retry and circuit-breaker patterns that improved feed reliability to 99.9% during volatile sessions.
- Built **React dashboards** for transaction surveillance alerts, translating complex risk rules into actionable UI workflows that accelerated analyst triage by 25%.
- Implemented algorithms supporting **income verification** and counterparty exposure checks within batch jobs processing 500K+ daily evaluations.
- Partnered with quantitative analysts to productionize **predictive risk models**, containerizing scoring services and documenting REST endpoints for downstream consumers.
- Reduced customer operations load by automating statement reconciliation workflows, saving 15+ analyst hours weekly through scripted validations and internal tooling.
- Contributed **GraphQL** read layers atop legacy REST services for mobile clients, simplifying client data fetching while preserving backward-compatible Java service boundaries.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Shipped **React** features for check-processing workflows from wireframes through production, applying **TypeScript** and modern ES6 module patterns across a remote-first squad.
- Extended **Java REST endpoints** for document upload and status tracking, adding validation middleware that decreased malformed request errors by 18%.
- Authored **SQL migrations** and reporting queries for audit tables, enabling operations teams to trace item lifecycle events without ad-hoc scripts.
- Collaborated on **Docker**-based local dev environments, shortening onboarding setup time for new interns from days to under four hours.
- Participated in **visual testing** pilots using snapshot comparisons, flagging unintended UI regressions before QA handoff on two release trains.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Delivered responsive **React** landing pages and donation flows for a nonprofit platform, improving mobile completion rates through tighter form validation and layout refinements.
- Integrated lightweight **REST API** clients for newsletter and CRM webhooks, enabling marketing staff to sync contacts without manual CSV uploads.
- Maintained **SQL-backed** content models and seed scripts, supporting rapid copy updates ahead of fundraising campaigns.
- Coordinated with stakeholders remotely to prioritize bug fixes and **UI polish**, clearing a 20-item backlog within the engagement window.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, React Native, TypeScript, ES6+, HTML5, CSS3, Interaction Design, Motion UI, A/B Testing, Visual Regression Testing

Backend: Java, REST APIs, GraphQL, Microservices, Transaction Evaluation, Risk Management Algorithms, Third-Party API Integration

Databases: PostgreSQL, Oracle, SQL, Schema Design, Migrations, Query Optimization, Analytical Queries

Cloud & DevOps: Docker, Kubernetes, CI/CD, Container Orchestration, Zero-Downtime Deployments

Monitoring & Tools: Visual Testing Automation, Feature Flags, Git, JUnit, Jest, Snapshot Testing

Workflow & PM: Agile, Cross-Functional Collaboration, Technical Mentorship, Stakeholder Communication, Independent Project Ownership